

The different GPUs

Unlike the CPU component, graphics portion is manufactured by multiple vendors, which gives companies the flexibility to pick and choose the GPU that goes best with the CPU in their SoC. While back in the day, the primary job of the graphics card was rendering 3D images and displaying these on the screen, today, GPUs are used for much more than just playing games and are as crucial as the CPU.

Today's OSes rely heavily on the GPU since the interface and all animations are rendered on the graphics chip. This is how you are able to get the buttery-smooth transition effects. It also comes in handy when watching HD videos. Just like its PC component, a graphics chip essentially has more cores than a CPU, thereby allowing it to distribute the load better and leaving the CPU free for other tasks.

Some of the most popular vendors for GPUs are:

ARM Mali. Mali is a series of GPUs produced by ARM. Unlike other GPUs, Mali chip does not actually



Fig. 2: ARM Cortex A15



Fig. 3: Need for Speed Most Wanted running on an iPhone

Smartphones: Their CPUs and GPUs

Smartphones	Chipset	CPU	GPU
HTC One M8	Qualcomm Snapdragon 801	Quad-core Krait 400	Adreno 330
Samsung Galaxy S5	Qualcomm Snapdragon 801	Quad-core Krait 400	Adreno 330
Samsung Galaxy Note 3	Qualcomm Snapdragon 800/ Exynos 5 octa 5420	Quad-core Krait 400/ quad-core Cortex A15 and quad-core Cortex A7	Adreno 330/ Mali-T628 MP6
Apple iPhone 5S	Apple A7	Dual-core Cyclone	Power VR G6430
Nokia Lumia 1520	Qualcomm Snapdragon 800	Quad-core Krait 400	Adreno 330
Samsung Galaxy Alpha	Exynos 5 octa 5430	Quad-core Cortex A15 and quad-core Cortex A7	Mali-T628 MP6
LG G3	Qualcomm Snapdragon 801	Quad-core Krait 400	Adreno 330
Micromax Canvas Knight	Mediatek 6592T	Octa-core Cortex A7	Mali-450MP4

have any display controller built-in to drive the LCD panel. Instead, it is just a pure 3D engine that does the job of rendering graphics into the memory. The most popular chip is Mali-400 MP, a quad-core GPU that is used in plenty of SoCs, like Samsung's Exynos and ST-Ericsson's NovaThor.

Qualcomm Adreno. Qualcomm changed the branding to Adreno from Imageon after a buyout by AMD. Adreno has since been used in Qualcomm's Snapdragon SoCs, their most popular ones being Adreno 205, Adreno 220 and Adreno 225.

Imagination PowerVR.

PowerVR GPUs are not manufactured by PowerVR themselves, but instead, they license their design and patents to other companies, like Texas Instruments, Intel, Samsung and Apple. Their PowerVR SGX series have been quite popular and feature SGX543XT, SGX544XT and SGX554XT.

Nvidia Geforce.

The name Geforce is synonymous with graphics cards ever since these were introduced, way back in 1999. Nvidia de-

veloped these graphics chips in-house and, till now, these only feature in their own Tegra SoCs. Their Tegra 3 SoC ups the ante by offering a 12-core GPU, whereas Tegra K1 offers 192 cores with shared architecture.

AnTuTu Benchmark comparison

AnTuTu Benchmarking is an application to assess the performance of smartphones and tablet PCs. The total score depends on various features, like CPU, GPU, RAM and IO. The multi-tasking score depends on the CPU and RAM. 2D and 3D graphics score depends on the GPU.

Future prospects

SoC manufacturers are using 32-bit SoC architecture (ARMv7) these days. After Apple set the ball rolling last year with the first 64-bit SoC used in iPhone 5S, all major manufacturers are likely to base their designs on the 64-bit architecture. The biggest advantage of ARMv8 is the support for more than 4GB RAM. Three major upcoming 64-bit SoCs are:

Qualcomm Snapdragon 610 and 615. The new Snapdragon 610 and 615 chipsets are 64-bit quad-core and octa-core chipsets, respectively.

Nvidia Tegra K1. The biggest advancement from Tegra4 is the inclusion of a Kepler based GPU with a staggering 192 shaders.

Mediatek MT6732. It features four Cortex-A53 CPUs capable of running up to 1.5GHz. It will also sport the new Mali-T760 GPU. ●